

Hedging Currency Exposures



Types of Currency Exposures

Currency risk pertains to adverse movements in exchange rate leading to loss in FX positions. There are three types of currency risk exposures

Transaction Exposures

- Commitment/Intention to receive or pay in foreign currency
- It's the forecasted revenues and expenses which are not in Balance sheet yet

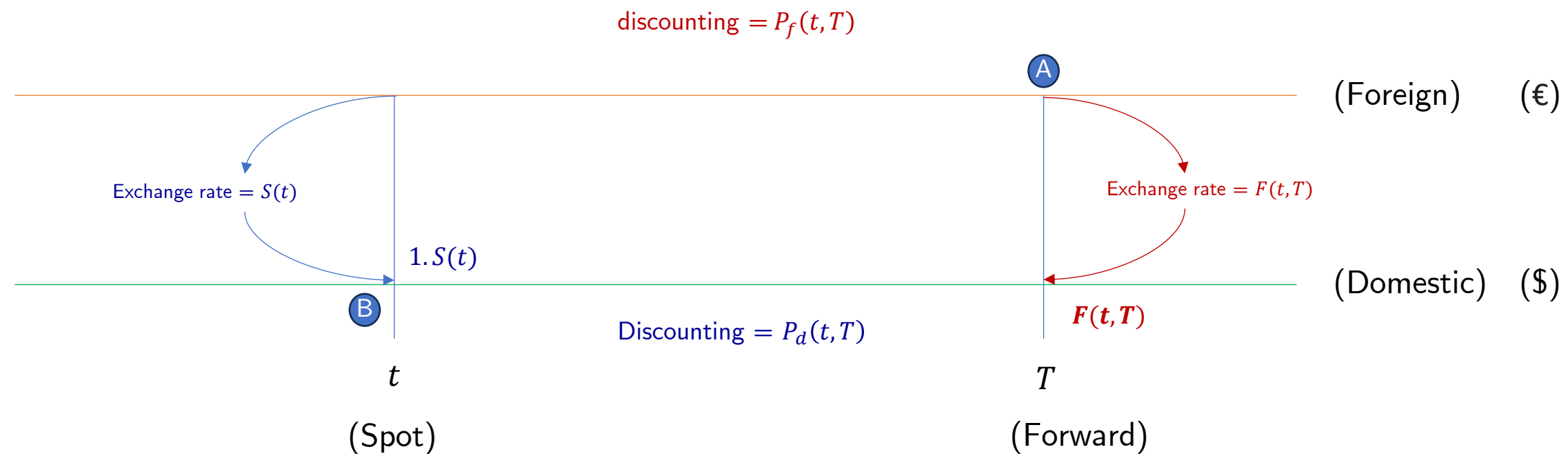
Translation Exposures

- Items on balance sheet denominated in foreign currency which need to be converted to domestic currency for reporting purposes
- These affect balance sheet and income statement

Economic Exposures

- Company's future receipts and payments where the transactions are not yet made but are exposed to exchange rate and economic environment of the country

Interest rate parity



$$F(t, T) = S(t) \frac{P_f(t, T)}{P_d(t, T)}$$

Forward Points

- ▶ Forward points is the basis points difference between Forward rate and Spot rate. So, when you add Forward points to the Spot rate, you get the Forward rate

$$F(t, T) = S(t) \frac{P_f(t, T)}{P_d(t, T)}$$

$$F(t, T) = S(t) e^{(r_d - r_f)(T - t)} \quad (\text{continuous compounding})$$

$$F(t, T) = S(t) \frac{(1 + i_d(T - t))}{(1 + i_f(T - t))} \quad (\text{simple compounding})$$

- ▶ Forward points

$$F(t, T) - S(t) = \frac{S(t) \left((i_d - i_f) \frac{D}{360} \right)}{\left(1 + i_f \frac{D}{360} \right)} \times 10000$$

Basic Hedging Instruments

▶ Forwards/Futures/Swaps Hedge

- For short term exposures use Forwards/Futures
- For long dated transactions use Swaps
- A long position in FX would require a short position in the hedging instrument

▶ Money Market hedge

- Take an equivalent loan in foreign currency as the receivable
- Convert to domestic currency in spot market and invest in money market fund in domestic currency
- At maturity, return the foreign loan and income on the security

▶ Option Hedge

- For FX receivable, buy FX puts
- For FX payable, buy FX calls

Simple Hedging Example



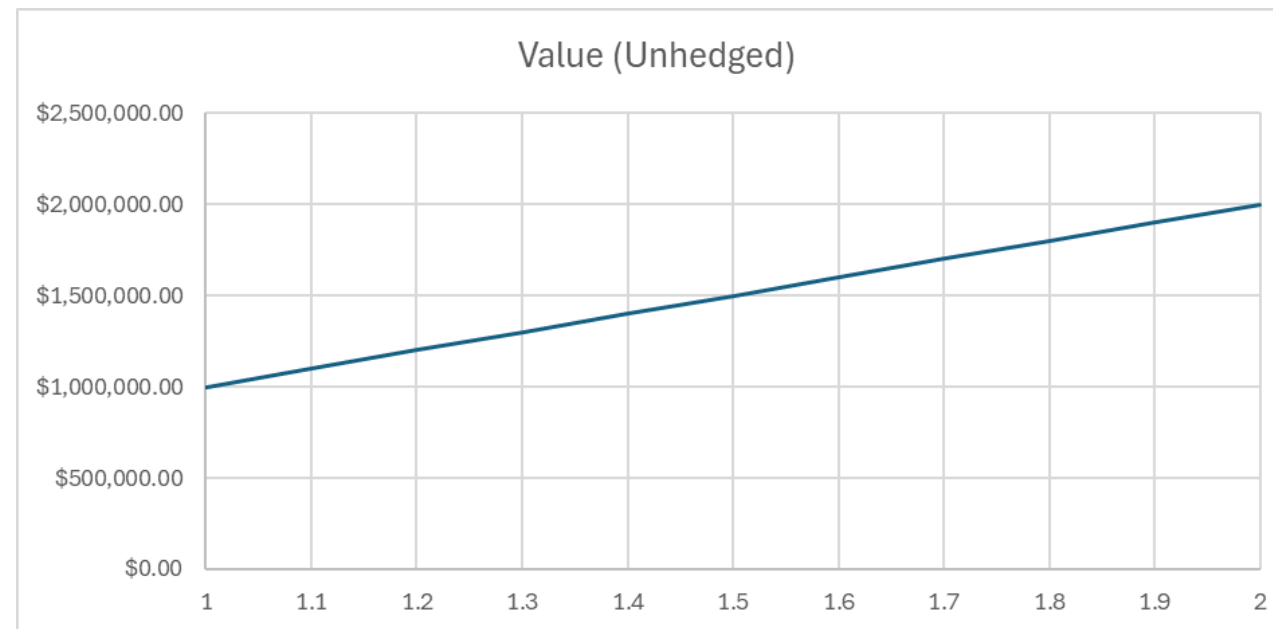
Suppose a USD bank has a EUR receivable 1 million € in one quarter..
Hedge the currency exposure using

- Forward Hedge
- Money Market Hedge
- Option Hedge

Unhedged Portfolio



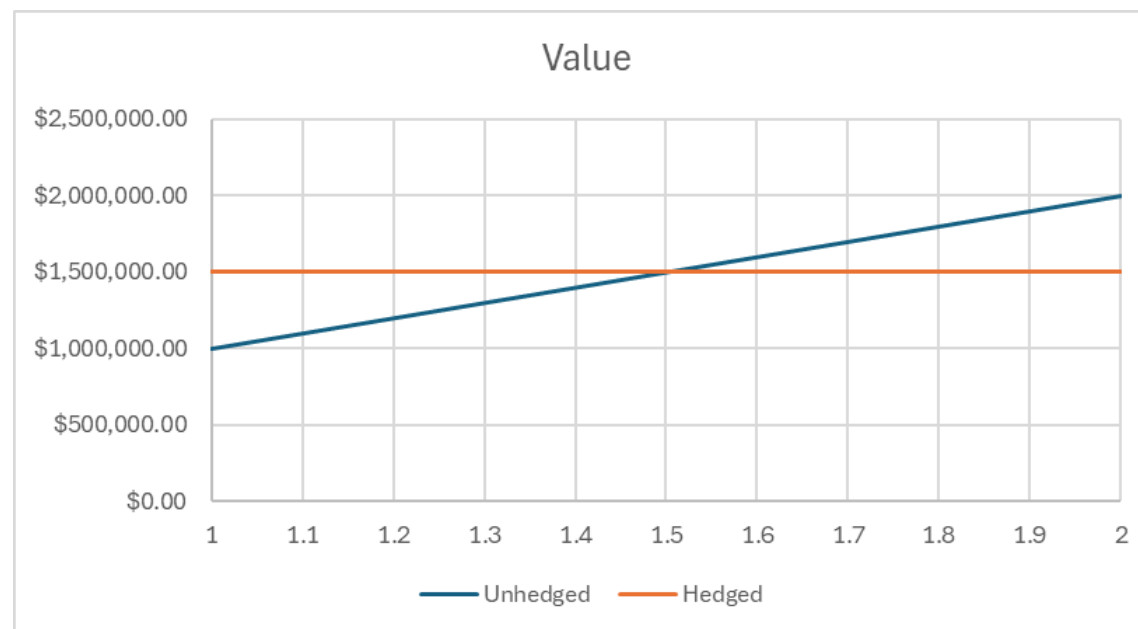
S(T)	V(T)
1	\$1,000,000.00
1.1	\$1,100,000.00
1.2	\$1,200,000.00
1.3	\$1,300,000.00
1.4	\$1,400,000.00
1.5	\$1,500,000.00
1.6	\$1,600,000.00
1.7	\$1,700,000.00
1.8	\$1,800,000.00
1.9	\$1,900,000.00
2	\$2,000,000.00



Forward Hedge

Since we are long on FX exposure, we need to enter a short Forward position

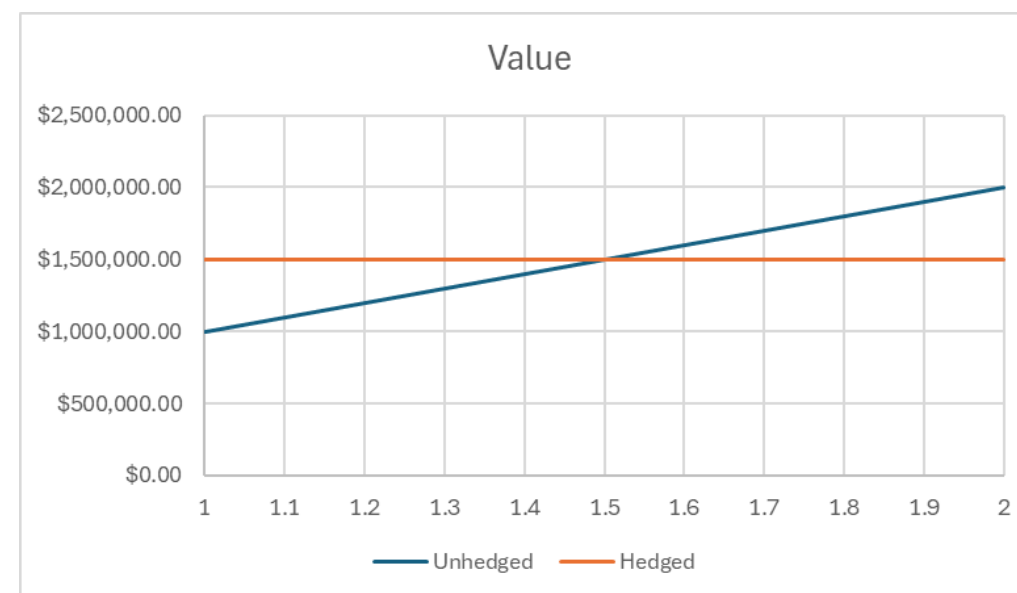
S(T)	Value (unhedged)	Forward Hedge
1	\$1,000,000.00	\$1,501,736.85
1.1	\$1,100,000.00	\$1,501,736.85
1.2	\$1,200,000.00	\$1,501,736.85
1.3	\$1,300,000.00	\$1,501,736.85
1.4	\$1,400,000.00	\$1,501,736.85
1.5	\$1,500,000.00	\$1,501,736.85
1.6	\$1,600,000.00	\$1,501,736.85
1.7	\$1,700,000.00	\$1,501,736.85
1.8	\$1,800,000.00	\$1,501,736.85
1.9	\$1,900,000.00	\$1,501,736.85
2	\$2,000,000.00	\$1,501,736.85



Money Market Hedge

- Get foreign loan such that the maturity payoff amount matches with the FX receivable
- Convert the loan into domestic currency in spot market
- Invest in domestic money market security

S(T)	Value (unhedged)	MM Hedge
1	\$1,000,000.00	\$1,501,736.85
1.1	\$1,100,000.00	\$1,501,736.85
1.2	\$1,200,000.00	\$1,501,736.85
1.3	\$1,300,000.00	\$1,501,736.85
1.4	\$1,400,000.00	\$1,501,736.85
1.5	\$1,500,000.00	\$1,501,736.85
1.6	\$1,600,000.00	\$1,501,736.85
1.7	\$1,700,000.00	\$1,501,736.85
1.8	\$1,800,000.00	\$1,501,736.85
1.9	\$1,900,000.00	\$1,501,736.85
2	\$2,000,000.00	\$1,501,736.85

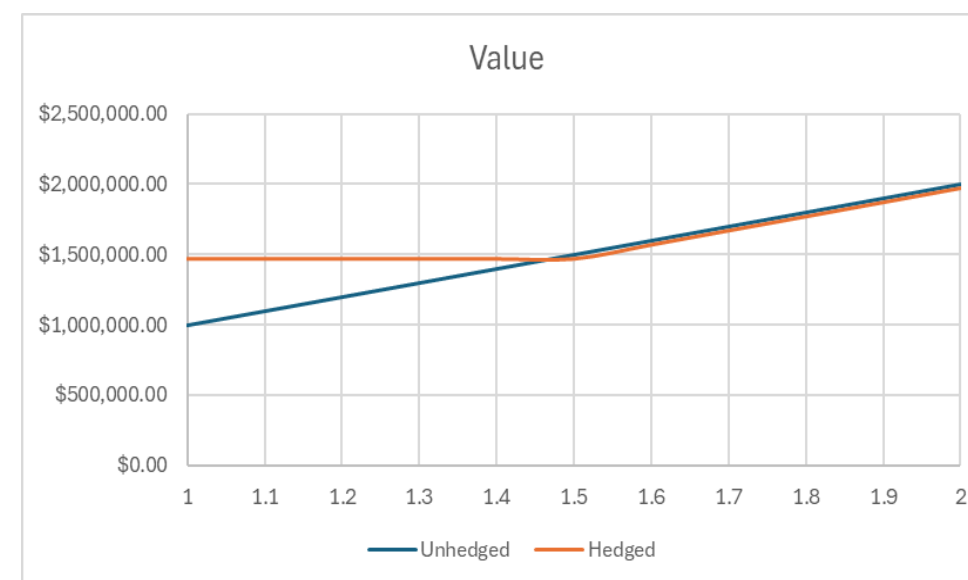


No difference in Forward Hedge vs Money Market hedge if IRP holds

Option Hedge

- Get into a long position on EUR put option

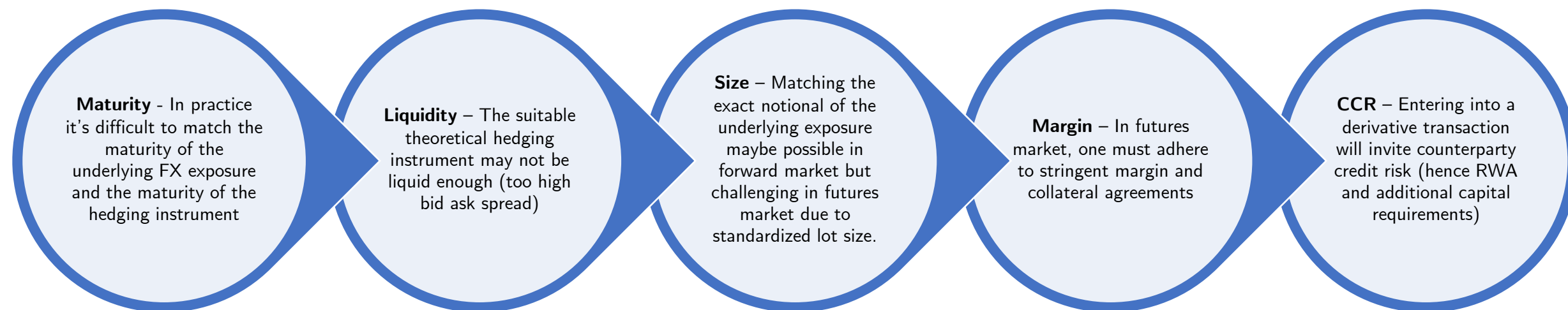
S(T)	Value (unhedged)	Forward (hedged)	Option (hedged)
1	\$1,000,000.00	\$1,471,254.62	\$1,471,254.62
1.1	\$1,100,000.00	\$1,471,254.62	\$1,471,254.62
1.2	\$1,200,000.00	\$1,471,254.62	\$1,471,254.62
1.3	\$1,300,000.00	\$1,471,254.62	\$1,471,254.62
1.4	\$1,400,000.00	\$1,471,254.62	\$1,471,254.62
1.5	\$1,500,000.00	\$1,471,254.62	\$1,471,254.62
1.6	\$1,600,000.00	\$1,571,254.62	\$1,571,254.62
1.7	\$1,700,000.00	\$1,671,254.62	\$1,671,254.62
1.8	\$1,800,000.00	\$1,771,254.62	\$1,771,254.62
1.9	\$1,900,000.00	\$1,871,254.62	\$1,871,254.62
2	\$2,000,000.00	\$1,971,254.62	\$1,971,254.62



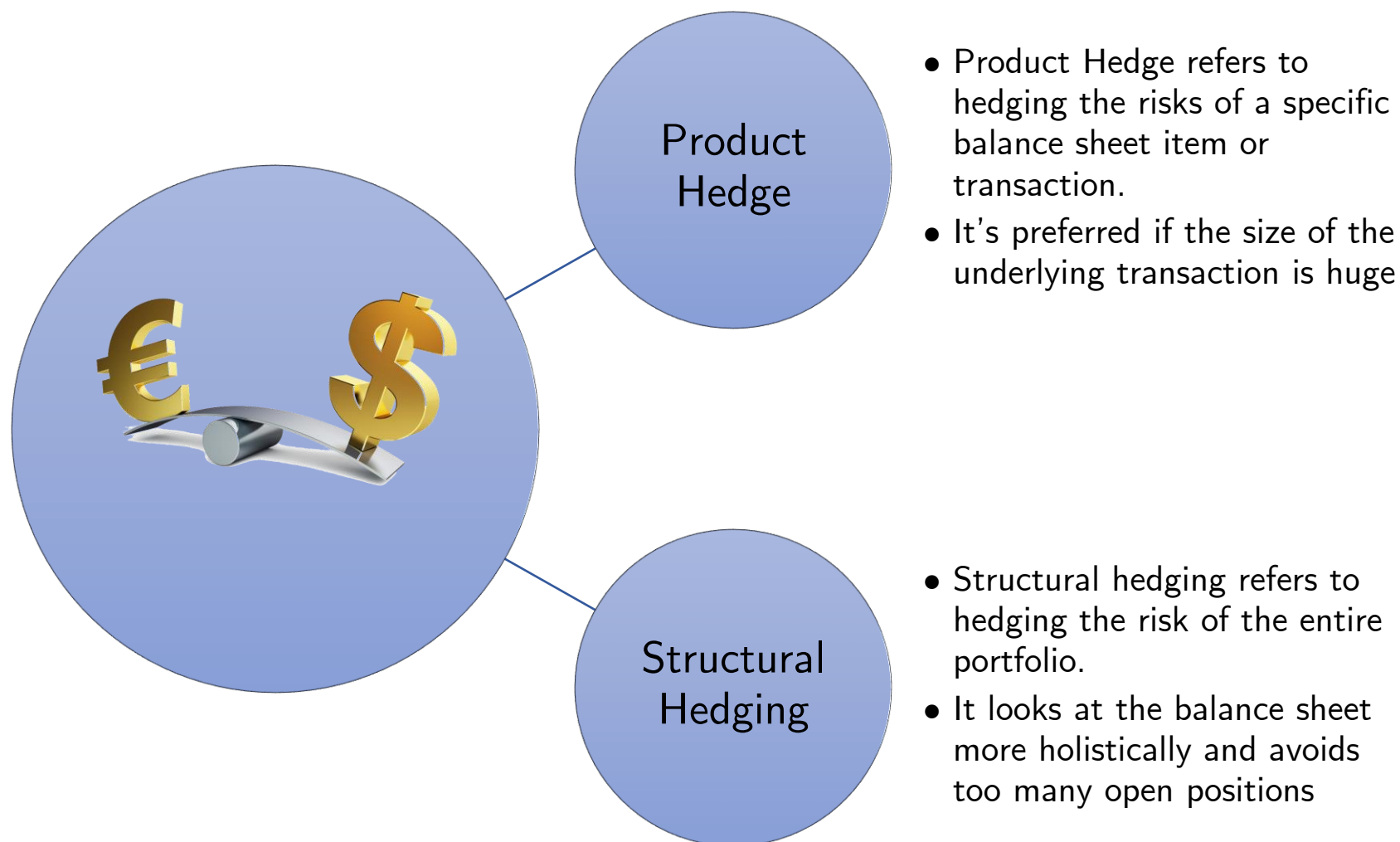
Best if there is a view on upside movement

The cost of additional upside potential is the option premium

Challenges in Practice



Broad Hedging strategies



Delta Hedge



- ▶ In the context of structural hedging, the popular approach is to delta hedge i.e., match the underlying delta with the delta of the hedging instrument.
- ▶ Key point to note is that while delta hedging FX, we also create a residual interest rate exposure. For short dated hedging instruments (e.g., 1 month/3 months), the interest rate exposure will be small. However, for long dated hedging instruments, the interest rate risk will be substantial.
- ▶ The preferred approach for structural hedging is to use short term FX forward and roll forward periodically
- ▶ In our example, we will use Forwards and Options

Delta Hedge

► Underlying Delta

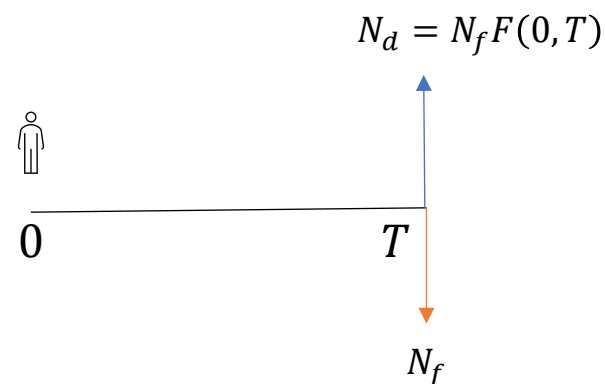


$$V = N_f e^{-r_f T} \cdot S$$

$$\delta_S = N_f e^{-r_f T}$$

$$\delta_{r_f} = -TV$$

► Forward Delta



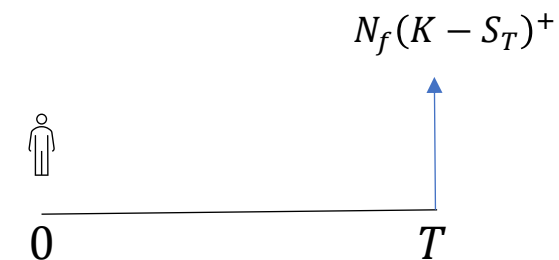
$$V = N_d e^{-r_d T} - N_f e^{-r_f T} S$$

$$\delta_S = -N_f e^{-r_f T}$$

$$\delta_{r_f} = T N_f e^{-r_f T} S$$

$$\delta_{r_d} = -T N_d e^{-r_d T}$$

► Option Delta



$$V(t) = N_f \left(K e^{-r_d(T-t)} \mathcal{N}(-d_2) - S_t e^{-r_f(T-t)} \mathcal{N}(-d_1) \right)$$

$$\delta_S = N_f (N(d_1) - 1) e^{-r_f T}$$